INFORMATION RETRIEVAL • FYP-BASED IMPLEMENTATION

Finding the right energy evidence

Applying 7 retrieval models to VoltMize

QUESTION

When a factory user describes a problem, which past incident or diagnostic record should appear first?

Course presentation • 12–13 October 2026

01 • FYP PROBLEM

VoltMize turns meter streams into actionable evidence

01

INGEST

Factory meter readings

02

DETECT

Abnormal consumption

03

EXPLAIN

Likely cause + context

04

ACT

Recommended response

IR retrieval problem

Retrieve the most relevant historical evidence for a natural-language factory-energy query.

02 • RETRIEVAL REQUIREMENT

One project question, two kinds of evidence

Element	Unstructured text	Structured text
Query	“compressor abnormal high energy night shift”	“weekend no production shutdown consumption”
Document / item	Technician incident report	Diagnostic JSON record
Result	Ranked reports	Ranked records / regions
Success	Relevant incidents in top 3	Correct diagnosis and action in top 3

Why two datasets? The models make different assumptions about document structure, so a fair demo gives each family the data shape it was designed for.

03 • MODEL MAP

From course concepts to VoltMize datasets

UNSTRUCTURED INCIDENT REPORTS

- Boolean — exact presence
- Vector Space — TF-IDF similarity
- BIM — probability from binary evidence
- BM25 — term frequency + rarity + length

STRUCTURED DIAGNOSTIC RECORDS

- Overlapped Lists — terms together in one region
- Non-Overlapped Lists — terms across regions
- Proximal Nodes — nearby fields / nodes

Not treated as retrieval models: clustering and classification are supporting organization tasks.

04 • EVALUATION

Five judged queries per data type • Top-3 evaluation

Ground truth was written before scoring: each query has an explicit relevant-document list.

PRECISION@3
hits ÷ 3
How clean are the first 3?

RECALL@3
hits ÷ all
How much did we recover?

F1@3
P + R
Precision + recall together

MRR
first-hit rank
How soon is first good hit?

nDCG@3
rank score
Rewards good ordering

Test discipline

Same tokenization • same queries • same relevance judgments • deterministic runs

This is a small classroom benchmark, so the scores demonstrate behavior—not production readiness.

DATA TYPE A

Unstructured text: 15 technician incident reports

Fictional but traceable to VoltMize events: energy anomalies, context, diagnosis, and maintenance action.

U01 • Compressor-1 night shift anomaly

“During the night shift Compressor-1 drew unusually high energy. Power stayed above 92 kW while production was idle. Inspect unload valve leakage and the pressure setpoint.”

Document = title + free-text note

WHY IT FITS

- Technicians naturally write prose
- Term frequency and rarity matter
- No reliable fields are required
- Search supports incident triage

5 queries • 15 documents • qrels included

QUERY → EVIDENCE → SCORE → RANK

Boolean Retrieval

OR selects a document when any query term is present; every selected document ties.

QUERY **compressor abnormal high energy night shift** — ANY MATCH → score 1

U01	✓ compressor ✓ high ✓ energy ✓ night ✓ shift	5 matches—but Boolean ignores count	1.000
U02	✓ shift	1 match receives same score	1.000
U03	✓ shift	1 match receives same score	1.000

TOP 3

1	U01	1.000
2	U02	1.000
3	U03	1.000

Tie shown in document-ID order.

WHY **Boolean answers MATCH / NO MATCH; it does not know that five matches are stronger than one.**

QUERY → EVIDENCE → SCORE → RANK

Vector Space Model

TF-IDF turns the query and reports into vectors; cosine similarity ranks their direction.

QUERY **compressor abnormal high energy night shift** TF-IDF → COSINE

U01	5 terms • compressor×2 • night×2 • shift×2	strong shared weighted direction	0.4365
U06	abnormal + compressor + shift	rare ‘abnormal’ adds high IDF	0.2413
U11	compressor×2 + night	fewer terms, still similar	0.2093

TOP 3

1	U01	0.4365
2	U06	0.2413
3	U11	0.2093

Largest cosine appears first.

WHY **Frequency and rarity shape the vectors; cosine compares the complete term-weight pattern.**

QUERY → EVIDENCE → SCORE → RANK

Binary Independence Model (BIM)

Each term is binary evidence; present rare terms contribute larger RSJ weights.

QUERY **compressor abnormal high energy night shift** PRESENT? × RARITY → SUM

U01	1 0 1 1 1 1 → five present terms	sum RSJ weights	4.7107
U06	1 1 0 0 0 1 → includes rare ‘abnormal’	rare evidence lifts score	4.4799
U11	1 0 0 0 1 0 → two present terms	repetition is ignored	2.9594

TOP 3

1	U01	4.7107
2	U06	4.4799
3	U11	2.9594

Binary vector order follows query terms.

WHY **BIM ranks by the strength of present evidence, not by how many times a word repeats.**

QUERY → EVIDENCE → SCORE → RANK

BM25

For every matching term: rarity × saturated frequency × length normalization.

QUERY **compressor abnormal high energy night shift** IDF × TF SATURATION × LENGTH

U01	5 terms • compressor×2 • night×2 • shift×2	rarity + repeat boost + length	7.9923
U06	abnormal + compressor + shift	rare terms; each occurs once	5.2178
U11	compressor×2 + night	repeat boost, lower coverage	4.0005

TOP 3

1	U01	7.9923
2	U06	5.2178
3	U11	4.0005

Scores use k1=1.5 and b=0.75.

WHY **BM25 rewards useful repetition but limits spam, then corrects for document length.**

CODE → FORMULA → RANK

Where BM25 scoring happens in the real code

The highlighted lines are the implementation that generated the scores on the previous slide.

unstructured_text/bm25/run.py

```
21  def search(query: str):
22      query_terms = tokenize(query)
23      scored = []
24      for doc_id, counts in doc_tf.items():
25          dl = sum(counts.values())
26          score = 0.0
27          for term in query_terms:
28              tf = counts[term]
29              if not tf:
30                  continue
31              idf = math.log(1 + (N - df[term] + 0.5) / (df[term] + 0.5))
32              score += idf * (tf * (k1 + 1)) / (tf + k1 * (1 - b + b * dl / avgdl))
33          if score > 0:
34              scored.append((doc_id, score))
35      return sorted(scored, key=lambda item: (-item[1], item[0]))
```

Highlighted lines are read directly from the implementation used for the measured results.

① TOKENIZE

Line 22 turns the query into terms.

② NORMALIZE

Line 25 measures document length.

③ SCORE

Lines 31–32 compute IDF and add each BM25 term score.

④ RANK

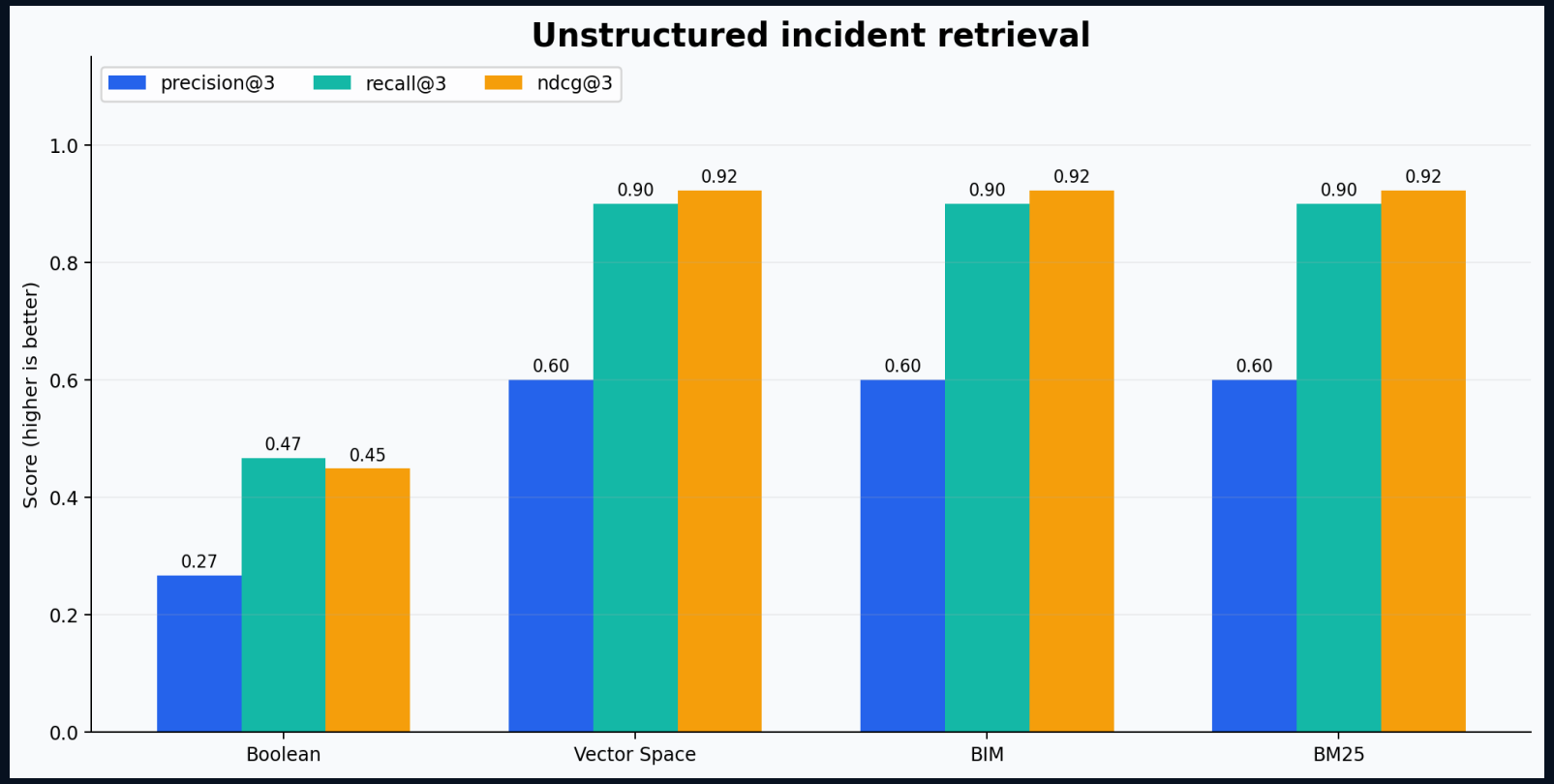
Line 35 sorts highest score first.

Same code → same scores → same top 3

UNSTRUCTURED COMPARISON

Ranking changes everything

Averages across UQ1–UQ5; higher is better.



RESULT

0.923

nDCG@3 for VSM, BIM & BM25

Selection

BM25

Tie broken by practical robustness: frequency saturation and length normalization.

DATA TYPE B

Structured text: 12 diagnostic JSON records

Each record preserves the stages VoltMize needs to explain an anomaly.

equipment	Compressor-1
summary	High energy during night shift
symptoms	94 kW • idle • cycling
meter_context	night • 488 kWh
probable_cause	unload valve leakage
action	inspect valve; reduce setpoint

WHY STRUCTURE MATTERS

- The same term can occur in different fields
- Cause and action may be separated
- Region co-occurrence changes meaning
- Tree distance can express closeness

QUERY → EVIDENCE → SCORE → RANK

Overlapped Lists

Count query terms inside each field; only the best single field becomes the score basis.

QUERY **night shift compressor high energy cause action** MAX MATCHES IN ONE FIELD

S01	summary: night + shift + high + energy	best field = 4 → 4 + 4/7	4.5714
S03	meter_context: energy + shift	best field = 2 → 2 + 2/7	2.2857
S06	meter_context: energy + night	best field = 2 → 2 + 2/7	2.2857

TOP 3

1	S01	4.5714
2	S03	2.2857
3	S06	2.2857

S02 is relevant, but its terms are split across fields.

WHY **This model rewards co-occurrence inside one region and ignores evidence scattered elsewhere.**

QUERY → EVIDENCE → SCORE → RANK

Non-Overlapped Lists

Merge unique matches from every field, then divide matched query terms by query length.

QUERY **night shift compressor high energy cause action** UNION ACROSS FIELDS ÷ 7

S01	equipment: compressor • summary: night shift high energy	5 unique terms ÷ 7	0.7143
S03	symptoms: high • context: energy shift	3 unique terms ÷ 7	0.4286
S02	equipment: compressor • symptoms/context: shift	2 unique terms ÷ 7	0.2857

TOP 3

1	S01	0.7143
2	S03	0.4286
3	S02	0.2857

Distributed evidence lets S02 enter the top 3.

WHY **Unlike overlap, a match can be assembled from equipment, symptoms, context, cause, and action.**

QUERY → EVIDENCE → SCORE → RANK

Proximal Nodes

Coverage finds the terms; node distance rewards matches that occur in the same or nearby fields.

QUERY **night shift compressor high energy cause action** 2×COVERAGE + Σ 1/(1+DISTANCE)

S01	equipment[0] ↔ summary[1] ↔ context[3]	5/7 coverage + many close pairs	9.4286
S03	high[2] ↔ energy,shift[3]	3/7 coverage + 1+.5+.5	2.8571
S06	energy[1,3] + night[3]	2/7 coverage + same-node pair	1.5714

TOP 3

1	S01	9.4286
2	S03	2.8571
3	S06	1.5714

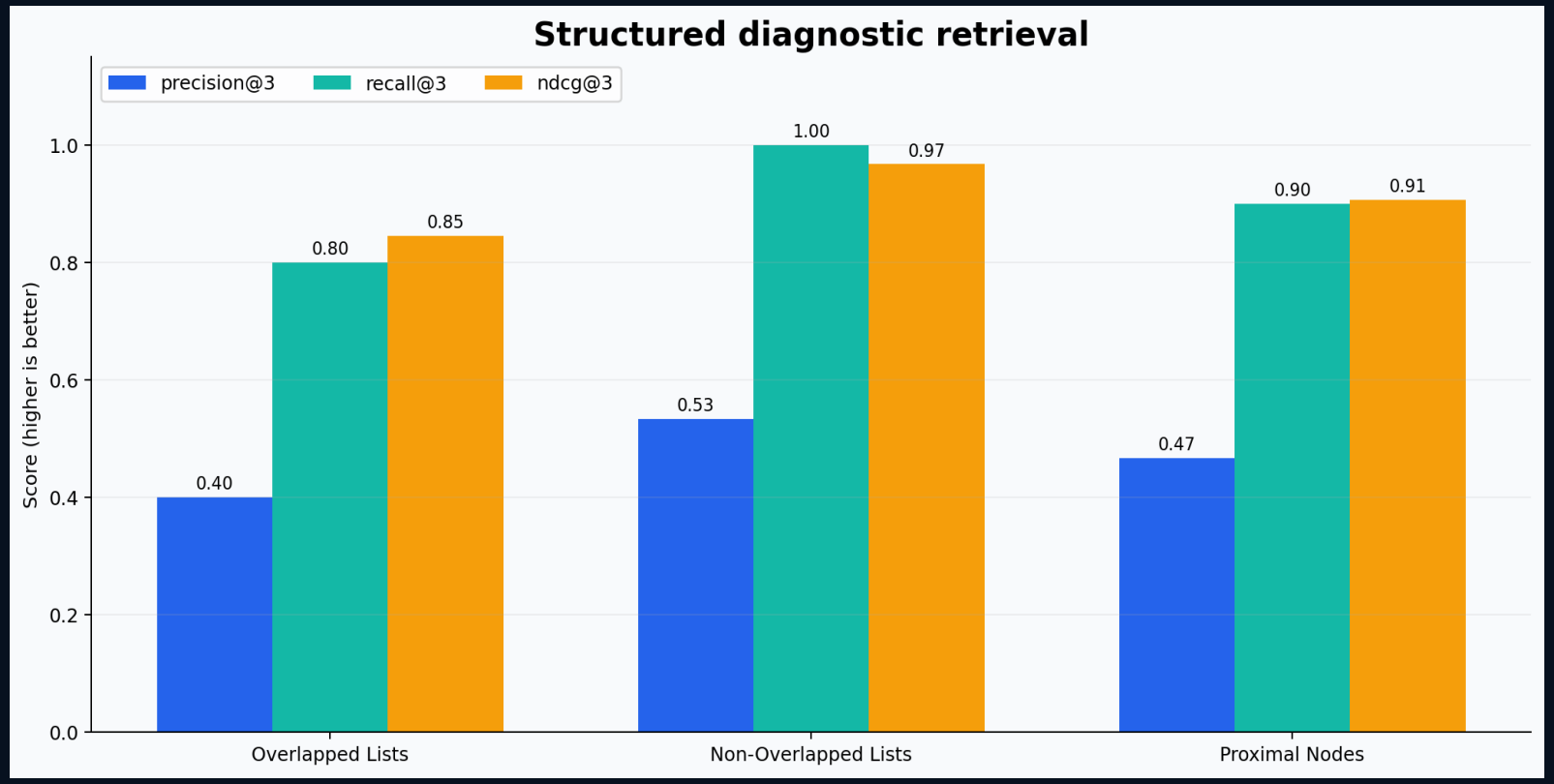
Node numbers follow the six-field record order.

WHY **Same node gives bonus 1; adjacent nodes .5; farther evidence contributes progressively less.**

STRUCTURED COMPARISON

Distributed evidence favors Non-Overlapped Lists

Averages across SQ1–SQ5; higher is better.



BEST nDCG@3

0.968

Non-Overlapped Lists

Why?

VoltMize queries mix equipment + symptom + context + action. Those clues live in different fields.

MODEL DECISION

There is no honest single winner across different data types

We select one model per evidence shape, then choose the measured overall leader for the final demo.

FREE-TEXT REPORTS

BM25

- Tied best measured nDCG@3: 0.923
- Most robust ranking assumptions
- Recommended incident-search baseline

STRUCTURED RECORDS • OVERALL LEADER

Non-Overlapped Lists

- Highest measured nDCG@3: 0.968
- Recall@3: 1.000
- Matches distributed diagnostic evidence

WHAT WE LEARNED

The model’s assumptions explain the results

Model behavior	Helps when…	Hurts when…
Exact presence	filters are strict	users describe problems loosely
Similarity / probability	documents are free text	field structure carries meaning
Same-region overlap	terms must co-occur	evidence spans fields
Across-region evidence	diagnosis is distributed	a phrase must stay together
Structural proximity	nearby nodes are meaningful	field order is only conventional

Core lesson

Choose a retrieval model because its assumptions match the document—not because its name sounds advanced.

APPENDIX • QUERY BANK

Ten VoltMize questions used for evaluation

UNSTRUCTURED

- UQ1 compressor abnormal high energy night shift
- UQ2 low power factor motor reactive energy
- UQ3 weekend idle consumption no production
- UQ4 temperature cooling chiller excess energy
- UQ5 missing meter readings communication

STRUCTURED

- SQ1 night shift compressor high energy cause action
- SQ2 low power factor reactive motor capacitor
- SQ3 weekend no production shutdown consumption
- SQ4 temperature chiller excess energy cause
- SQ5 missing meter readings communication action

CONCLUSION

From meter data to the right evidence

We connected course theory to a working, measured retrieval component for VoltMize.

FINAL TAKEAWAY

BM25 for incident prose. Non-Overlapped Lists for structured diagnosis.

Retrieve → rank → evaluate → explain

Questions?